

Mathematical Induction

Q.1: Show that $1^2 + 3^2 + 5^2 + \dots + (2n+1)^2 = \frac{(n+1)(2n+1)(2n+3)}{3}$

Where n is a non-negative integers

Ans: Non-negative means $n=0$ is the satisfying point.

Let $P(n)$ is a propositional statement.

Basic step: for $n=0$, $1^2 = \frac{(1)(1)(3)}{3} = 1$, it is true.

Inductive step: Let $n=k$, it is true then

$$1^2 + 3^2 + 5^2 + \dots + (2k+1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}$$

Now add $(2k+3)^2$ on both sides

$$1^2 + 3^2 + 5^2 + \dots + (2k+1)^2 + (2k+3)^2 = \frac{(k+1)(2k+1)(2k+3)}{3} + (2k+3)^2$$

$$= (2k+3) \left[\frac{(k+1)(2k+1) + 3(2k+3)}{3} \right]$$

$$= \frac{(2k+3)}{3} [2k^2 + 3k + 1 + 6k + 9]$$

$$= \frac{(2k+3)}{3} [2k^2 + 9k + 10]$$

$$= \frac{(2k+3)}{3} [2k^2 + 5k + 4k + 10]$$

$$= \frac{(2k+3)}{3} [2k(k+2) + 5(k+2)]$$

$$= \frac{(k+2)(2k+3)(2k+5)}{3} \Rightarrow \text{it is true for } P(k+1)$$

$\Rightarrow$ It is true for all n

Q.2: Show that $2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2 \cdot 7^n = \frac{1 - (-7)^{n+1}}{4}$

where n is a non-negative integer.

Ans:

Basic Step: for $n=0$, $2 = \frac{1 - (-7)^1}{4} = \frac{8}{4} = 2 \Rightarrow$ it is true

Inductive Step: Let $n=k$

$$P(k): 2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2 \cdot 7^k = \frac{1 - (-7)^{k+1}}{4}$$

For $P(k+1)$, add $2 \cdot 7^{k+1}$ both sides

$$2 - 2 \cdot 7 + 2 \cdot 7^2 - \dots + 2 \cdot 7^k + 2 \cdot 7^{k+1} = \frac{1 - (-7)^{k+1}}{4} + 2 \cdot 7^{k+1}$$

$$= \frac{1 - (-7)^{k+1} + 4 \times 2 \cdot 7^{k+1}}{4} = \frac{1 + 7(-7)^{k+1}}{4}$$

$$= \frac{1 - (-7)^{k+2}}{4} = \frac{1 - (-7)^{(k+1)+1}}{4}$$

$P(k+1)$ is also true.

$\Rightarrow$ Hence it is true for all n .

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